



Modelling Financial Markets

Seminar 11: Continuous-Time Models — Hands-on



Daniel Traian PELE

Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ CI: Confidence Interval
- ▣ CIR: Cox–Ingersoll–Ross (model)
- ▣ CKLS: Chan–Karolyi–Longstaff–Sanders (model)
- ▣ DTB3: 3-Month Treasury Bill: secondary market rate (FRED series code)
- ▣ FRED: Federal Reserve Economic Data
- ▣ GBM: Geometric Brownian Motion
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ JB: Jarque–Bera (test)
- ▣ LR: Likelihood Ratio (test)
- ▣ MCS: Model Confidence Set
- ▣ MLE: Maximum Likelihood Estimation
- ▣ MSE: Mean Squared Error
- ▣ NLLS: Non-Linear Least Squares
- ▣ OLS: Ordinary Least Squares
- ▣ OU: Ornstein–Uhlenbeck (process)
- ▣ PDE: Partial Differential Equation
- ▣ QLIKE: Quasi-LIKElihood (loss function)
- ▣ QQ: Quantile–Quantile (plot)
- ▣ RFSV: Rough Fractional Stochastic Volatility (model)
- ▣ RV: Realised Variance
- ▣ SDE: Stochastic Differential Equation
- ▣ US: United States

Acronyms used in this material (2/2)

- **VIX: CBOE Volatility Index**

Today's Question and Required Outputs

- Question of the seminar: **what can daily data tell us about a continuous-time model, and what can they not?**
- Route
 - ▶ What you need for today: five primer slides on Brownian motion, Itô, GBM, mean reversion and jumps ([Chapter 11](#) comes after the seminar)
 - ▶ A9, A4, A7 on paper: from an AR(1) to an OU process, Itô integrals, Merton cumulants
 - ▶ B2: GBM on the S&P 500 with robust uncertainty; B4: the bias of the mean-reversion speed
 - ▶ C2: find the errors in an AI answer; B3 at home; B1 and B6 after the lecture
- Required outputs (checklist on the last core slide): the A answers, one table and one figure per B exercise, the corrected C2 code, one interpretation sentence per exercise
- Notebook for today: [open the seminar notebook in Colab](#); each task names its notebook section

Exercise Map

Exercise	Question	Type	Model
A9	how precise is κ from an AR(1)?	Solved	–
A4	what does the extra term in $\int W dW$ measure?	Solved	–
A7	how non-Normal do five jumps a year make daily returns?	Solved	–
B2	is the S&P 500 a GBM, and how well do we know σ and m ?	Solved	–
B4	do interest rates really mean-revert?	Solved	–
C2	what is wrong in an AI answer about GBM?	Proposed	A4, B2
B3	BET and Bitcoin: same tests as B2 (at home)	Proposed	B2
After the lecture	B1: how fast do Euler and Milstein converge? B6: do the data reject GBM in favour of jumps?	Solved	–
Extensions	A1, A2, A3, A5, A6, A8; B4, B6 and B8 Extended, B5, B7, B8, B9; C1, C3 (optional, after the seminar)	both	on each task

- **[Solved]**: full solution in the slides and the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- Part A: derivations on paper; Part B: estimation and inference on data, each ending with an interpretation; Part C: open questions and AI critique

Data Used

Series	Symbol, field	Calendar, cleaning	Sample	Returns
S&P 500	GSPC.INDX, close	weekdays; stale quote = unchanged close and high = low	1990 – 2026	$n = 9\,245$
BET	BET, close	weekdays; the BET series has no daily high/low, so every unchanged close counts as stale (none occurs)	2000 – 2026	$n = 6\,670$
Bitcoin	BTC-USD.CC, close	7 days a week, $\Delta t = 1/365$	2014 – 2026	$n = 4\,384$
VIX	VIX.INDX, close	weekdays	1990 – 2026	$n = 9\,283$ (9 284 levels)
DTB3	FRED, percent /100	no filter	1954 – 2026	$n = 18\,170$ levels

- Daily prices of the course data, one series per asset (field used: close); end date 18 September 2026
- Log returns $r_t = \ln P_t - \ln P_{t-1}$ on each series' own calendar; genuine zero returns (unchanged close on a day with trading) are kept; extensions also use SPY 5-minute bars and the [Chapter 9](#) realised variance

Setup: Run and Verify

```
# notebook section "Data" defines
# log_returns(key), load_vix(), read_fred(id)
rets = {k: log_returns(k)
        for k in ['sp500', 'bet', 'btc']}
for k, r in rets.items():
    print(k, len(r), r.index[0].date(),
          r.index[-1].date())
tb = read_fred('DTB3') / 100
print('DTB3', len(tb), tb.index[0].date())
print('VIX', len(load_vix()))
```

Expected output

	<i>n</i>	first date
sp500	9 245	1990-01-03
bet	6 670	2000-01-06
btc	4 384	2014-09-18
DTB3 (levels)	18 170	1954-01-04
VIX (levels)	9 284	1990-01-02

- Expected output
- All series end on 2026-09-18; returns start one day after the first price
- If a count differs, stop and check the symbol, the field and the start date

Notation Used in Every Exercise

- ▣ W_t : standard Brownian motion; P : real-world measure, Q : risk-neutral measure; Φ : standard Normal distribution function
- ▣ Itô's lemma: $dX = a dt + b dW \Rightarrow df(X) = (af' + \frac{1}{2}b^2f'') dt + bf' dW$; moments $\mathbb{E}[W_t^2] = t$, $\mathbb{E}[W_t^4] = 3t^2$, $\mathbb{E}[e^{aW_t}] = e^{a^2t/2}$
- ▣ GBM: $\ln(S_T/S_0) \sim N(mT, \sigma^2 T)$, log drift $m = \mu - \sigma^2/2$; $\mathbb{E}[S_T] = S_0 e^{\mu T}$
- ▣ OU: $dx = \kappa(\theta - x) dt + \sigma dW$; sampled at step Δ : AR(1) with $b = e^{-\kappa\Delta}$, $a = \theta(1 - b)$; half-life $\ln 2/\kappa$
- ▣ Merton: $r_\Delta = m\Delta + \sigma\sqrt{\Delta}Z + \sum_{j \leq N_\Delta} Y_j$, $N_\Delta \sim \text{Poisson}(\lambda\Delta)$, $Y_j \sim N(\mu_J, \sigma_J^2)$; Heston: $dv = \kappa(\theta - v) dt + \xi\sqrt{v} dW^{(2)}$, $\text{Corr} = \rho$
- ▣ Units: time in years; drifts and volatilities per year; standard errors (s.e.) of percentages in percentage points; every estimate with its s.e. or confidence interval (CI)

What You Need for Today (1/5): Brownian Motion

- Chapter 11 comes after this seminar: these five slides are the minimum for today's core tasks; depth and proofs in the lecture
- Standard Brownian motion W_t : $W_0 = 0$, continuous paths, independent increments $W_{t+\Delta} - W_t \sim N(0, \Delta)$; the limit of a random walk with steps $\pm\sqrt{\Delta}$
 - ▶ The standard deviation of an increment is $\sqrt{\Delta}$, not Δ : one trading day, $\Delta = 1/252$, has s.d. 0.063; a simulated step is $\sigma\sqrt{\Delta}Z$ with $Z \sim N(0, 1)$
- Quadratic variation: over a grid of $[0, T]$ with step Δ , $\sum_i (\Delta W_i)^2$ has mean T and variance $2T\Delta \rightarrow 0$, so it converges to T
 - ▶ Shorthand: $(dW)^2 = dt$, $dt dW = 0$, $(dt)^2 = 0$; a smooth path has quadratic variation 0; realised variance (Chapter 9) estimates the quadratic variation of the log price
- Tiny example: 252 daily increments over one year: $\mathbb{E} \sum (\Delta W_i)^2 = 252 \times 1/252 = 1$, with s.d. $\sqrt{2/252} = 0.089$

What You Need for Today (2/5): the Itô Integral and Itô's Lemma

- Itô integral **Itô (1944)**: $\int_0^T f_t dW_t = \lim \sum_i f_{t_i} (W_{t_{i+1}} - W_{t_i})$, the integrand taken at the **left** point, known before the increment, like a position chosen before the price moves
 - ▶ Mean 0; Itô isometry $\text{Var} \left(\int_0^T f_t dW_t \right) = \int_0^T \mathbb{E}[f_t^2] dt$; the Stratonovich integral takes the mid-point instead and has a different mean (A4)
- Itô's lemma: if $dX = a dt + b dW$, then $df(X) = \left(af' + \frac{1}{2} b^2 f'' \right) dt + bf' dW$
 - ▶ Intuition: a second-order Taylor expansion in which $(dX)^2 = b^2 dt$ does not vanish; the ordinary chain rule drops exactly the term $\frac{1}{2} b^2 f'' dt$
- Tiny example: $f(x) = x^2$ and $X = W$ ($a = 0$, $b = 1$): $d(W^2) = 2W dW + dt$; the extra dt is the quadratic variation of the previous slide
- Moments you will need: $\mathbb{E}[W_t^2] = t$, $\mathbb{E}[W_t^4] = 3t^2$, $\mathbb{E}[e^{aW_t}] = e^{a^2 t/2}$

What You Need for Today (3/5): Geometric Brownian Motion

- GBM Samuelson (1965): $dS = \mu S dt + \sigma S dW$: the return over dt has mean μdt and variance $\sigma^2 dt$
- Itô with $f = \ln S$ ($f' = 1/S$, $f'' = -1/S^2$): $d \ln S = (\mu - \sigma^2/2) dt + \sigma dW$, so $\ln(S_T/S_0) \sim N(mT, \sigma^2 T)$ with the log drift $m = \mu - \sigma^2/2$
 - ▶ Mean $\mathbb{E}[S_T] = S_0 e^{\mu T}$, median $S_0 e^{mT}$, lower: a few large outcomes pull the mean up
- Tiny example: $\mu = 10\%$, $\sigma = 30\%$: $m = 0.10 - 0.045 = 5.5\%$; $P(S_1 < S_0) = \Phi(-m/\sigma) = \Phi(-0.18) = 0.43$
- Estimation from n daily log returns with step Δ over $T = n\Delta$ years: $\hat{\sigma}^2 = \text{sample variance}/\Delta$, $\hat{m} = \text{sample mean}/\Delta$
 - ▶ $\text{s.e.}(\hat{\sigma}) \approx \hat{\sigma}/\sqrt{2n}$ shrinks with the number of observations, $\text{s.e.}(\hat{m}) = \hat{\sigma}/\sqrt{T}$ only with the calendar span (B2)

What You Need for Today (4/5): Mean Reversion

- Ornstein–Uhlenbeck (OU) process [Uhlenbeck & Ornstein \(1930\)](#): $dx = \kappa(\theta - x)dt + \sigma dW$: pulled towards the level θ at speed $\kappa > 0$, shocked with volatility σ
 - ▶ Vasicek [Vasicek \(1977\)](#): the OU process as a model of the short interest rate
- Sampled every Δ years, an OU process is **exactly** an AR(1): $x_{t+\Delta} = a + b x_t + \varepsilon_t$
 - ▶ $b = e^{-\kappa\Delta}$, $a = \theta(1 - b)$, $\text{Var}(\varepsilon_t) = \sigma^2(1 - b^2)/(2\kappa)$; half-life $\ln 2/\kappa$: the time for an expected deviation from θ to halve
- Tiny example: monthly data ($\Delta = 1/12$) with $\hat{b} = 0.95$: $\hat{\kappa} = -12 \ln 0.95 = 0.62$ a year, half-life $\ln 2/0.62 = 1.1$ years
- In short samples the OLS $\hat{b}$ of an autoregression is biased downwards, so $\hat{\kappa}$ is biased upwards: a random walk can look mean-reverting (B4)

What You Need for Today (5/5): Jumps in the Merton Model

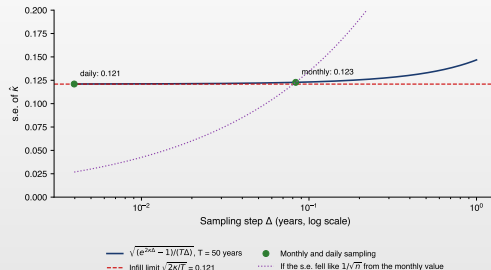
- Merton [Merton \(1976\)](#): GBM plus jumps that arrive as a Poisson process with intensity λ per year; each jump adds $Y \sim N(\mu_J, \sigma_J^2)$ to the log price
 - ▶ Over a day the number of jumps is $N_\Delta \sim \text{Poisson}(\lambda\Delta)$; $\lambda = 10$ gives a jump on about $10/252 = 4\%$ of days
 - ▶ The daily density is a Poisson mixture of Normal densities: given k jumps, $r_\Delta \sim N(m\Delta + k\mu_J, \sigma^2\Delta + k\sigma_J^2)$
- Cumulants: $K(u) = \ln \mathbb{E}[e^{ur}]$, $c_k = K^{(k)}(0)$; c_1 mean, c_2 variance, skewness $c_3/c_2^{3/2}$, excess kurtosis c_4/c_2^2 ; for independent parts the cumulants add
 - ▶ The diffusion contributes only to c_1 and c_2 ; skewness and fat tails come from the jumps (A7)
- If every jump were observed, $N_T \sim \text{Poisson}(\lambda T)$ and $\hat{\lambda} = N_T/T$ (the Poisson MLE); in returns the jumps are hidden among the diffusive moves

A9 [Solved]: From an AR(1) to an OU Process — Task

- ▣ **Question:** how precisely does a 50-year monthly sample pin down the speed of mean reversion?
- ▣ **Inputs:** an AR(1) estimated on $n = 600$ monthly transitions ($T = n\Delta = 50$ years, $\Delta = 1/12$) of a rate in decimals: $x_{t+1} = 0.0012 + 0.97x_t + \varepsilon_t$, residual s.d. $\hat{s}_\varepsilon = 0.0025$; the rate is an exactly sampled OU process, so $b = e^{-\kappa\Delta}$, $a = \theta(1 - b)$, $s^2 = \sigma^2(1 - b^2)/(2\kappa)$
- ▣ **Tasks**
 1. Invert the three relations to obtain κ , θ , σ and the half-life $\ln 2/\kappa$.
 2. Use the stationary large-sample approximation $\text{Var}(\hat{b}) \approx (1 - b^2)/n$ and the delta method to compute the s.e. of $\hat{\kappa}$.
 3. Build a 95% interval for the half-life by transforming the interval for κ (valid only if both ends are positive).
 4. Keep $T = 50$ years but sample daily ($\Delta = 1/252$): compute b , the s.e. of $\hat{\kappa}$ and its limit $\sqrt{2\kappa/T}$.
- ▣ **Report:** a table with parameter, formula, value and s.e.; the half-life interval; the daily s.e.
- ▣ **Interpretation:** does daily sampling make κ more precise?
- ▣ **Notebook:** section A9

A9 [Solved]: Solution — Mapping, Delta Method and Sampling Frequency

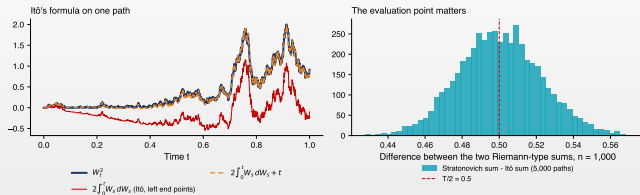
- $\kappa = -\ln b/\Delta = -12 \ln 0.97 = 0.366$;
 $\theta = a/(1 - b) = 0.0012/0.03 = 4.0\%$;
 $\sigma = s\sqrt{2\kappa/(1 - b^2)} = 0.88\%$; half-life $\ln 2/\kappa = 1.90$ years
 - $\text{s.e.}(\hat{b}) = \sqrt{(1 - 0.97^2)/600} = 0.0099$; $\partial\kappa/\partial b = -1/(b\Delta)$,
 so $\text{s.e.}(\hat{\kappa}) = 0.0099/(0.97/12) = 0.123$
 - $\kappa \in [0.125, 0.606]$, both ends positive, so the half-life lies in
 $[\ln 2/0.606, \ln 2/0.125] = [1.14, 5.55]$ years: asymmetric
 around 1.90
 - Daily: $b = e^{-\kappa/252} = 0.99855$, $n = 12\,600$, s.e. 0.121; in
 general $\text{Var}(\hat{\kappa}) \approx (e^{2\kappa\Delta} - 1)/(T\Delta) \rightarrow 2\kappa/T$, limit 0.121
 (chart)
- ☐ **Interpretation:** 21 times more observations give the same
 precision: κ is identified by calendar time, not by the number
 of points; if the interval for κ reached zero, the half-life
 would be unbounded



A4 [Solved]: Itô Integrals — Task

- ▣ **Question:** what does the extra term in $\int_0^T W \, dW$ measure, and why is its mean zero?
- ▣ **Inputs:** a standard Brownian motion W on $[0, T]$, $T = 1$ for the numbers; $\mathbb{E}[W_t^2] = t$, $\mathbb{E}[W_t^4] = 3t^2$; no data
- ▣ **Tasks**
 1. Apply Itô's lemma to $f(x) = x^2$ and show that $\int_0^T W \, dW = \frac{1}{2}(W_T^2 - T)$.
 2. Compute the mean and the variance of this integral and check the variance with the Itô isometry.
 3. The Stratonovich (mid-point) integral equals $\frac{1}{2}W_T^2$: compute its mean.
 4. Apply Itô's lemma to W_t^3 and deduce the mean and the variance of $\int_0^1 W^2 \, dW$.
- ▣ **Report:** a table with integral, mean and variance for $\int W \, dW$, the Stratonovich integral and $\int W^2 \, dW$
- ▣ **Interpretation:** why must an Itô integral have mean zero, while the Stratonovich integral does not?
- ▣ Notebook: section A4

A4 [Solved]: Solution — Identity, Moments and a Simulation Check



1. $d(W^2) = 2W dW + dt$; integrating from 0 to T : $W_T^2 = 2 \int_0^T W dW + T$, hence $\int_0^T W dW = \frac{1}{2}(W_T^2 - T)$
2. Mean $\frac{1}{2}(\mathbb{E}[W_1^2] - 1) = 0$; variance $\frac{1}{4}\text{Var}(W_1^2) = \frac{1}{4}(3 - 1) = 0.5$; isometry: $\int_0^1 \mathbb{E}[W_t^2] dt = \int_0^1 t dt = 0.5$
3. Stratonovich: mean $\frac{1}{2}\mathbb{E}[W_1^2] = 0.5 \neq 0$
4. $d(W^3) = 3W^2 dW + 3W dt$; $\mathbb{E} \int_0^1 W^2 dW = 0$ and $\text{Var} = \int_0^1 \mathbb{E}[W_t^4] dt = \int_0^1 3t^2 dt = 1$
5. Chart (lecture simulation): left, $2 \int_0^t W dW + t$ lies on W_t^2 ; right, Stratonovich minus Itô over 5,000 paths has mean 0.500; Itô sums have variance 0.520

□ **Interpretation:** the Itô integrand is fixed before the increment, so a strategy cannot anticipate the move; the mid-point rule uses the next move and earns $T/2$ on average: that is the quadratic variation

A7 [Solved]: Merton: Cumulants and the Information on λ — Task

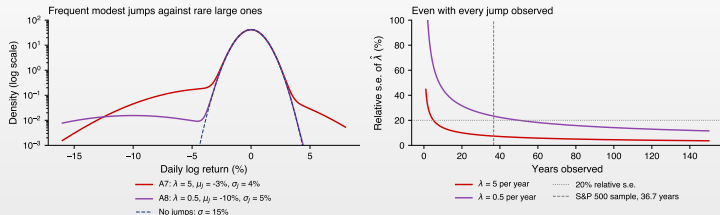
- **Question:** how non-Normal do five modest jumps a year make daily returns, and how well can λ be learnt?
- **Inputs:** $r_\Delta = m\Delta + \sigma\sqrt{\Delta}Z + \sum_{j=1}^{N_\Delta} Y_j$ with $Z \sim N(0, 1)$, $N_\Delta \sim \text{Poisson}(\lambda\Delta)$, $Y_j \sim N(\mu_J, \sigma_J^2)$, all independent; $\sigma = 15\%$, $\lambda = 5$ per year, $\mu_J = -3\%$, $\sigma_J = 4\%$, $\Delta = 1/252$; for step 4, $T = 36.7$ years
- **Tasks**
 1. Write the cumulant generating function $K(u) = \ln \mathbb{E}[e^{u r_\Delta}]$ and show that the cumulants (denoted c_k , to avoid confusion with the OU speed κ) are $c_k = \lambda\Delta \mathbb{E}[Y^k]$ for $k \geq 3$.
 2. Compute c_2 , split into its diffusion and jump parts, then c_3 and c_4 .
 3. Compute the skewness $c_3/c_2^{3/2}$ and the excess kurtosis c_4/c_2^2 .
 4. Assume every jump is observed: derive the Fisher information of the Poisson count N_T and the s.e. of $\hat{\lambda} = N_T/T$ at $\lambda = 5$; repeat at the lecture MLE $\hat{\lambda} = 101.9$ and compare with its MLE s.e. 10.5.
- **Report:** a table with the diffusion and jump parts of c_2 , c_3 , c_4 , skewness, excess kurtosis; the two standard errors
- **Interpretation:** why is the MLE s.e. much larger than the s.e. with observed jumps?
- Notebook: section A7

A7 [Solved]: Solution (1/2) — Cumulants

1. $K(u) = u m \Delta + \frac{1}{2} u^2 \sigma^2 \Delta + \lambda \Delta \left(\mathbb{E}[e^{uY}] - 1 \right)$; the k -th derivative at 0 gives $c_k = \lambda \Delta \mathbb{E}[Y^k]$ for $k \geq 3$
2. $\mathbb{E}[Y^2] = \mu_J^2 + \sigma_J^2$, $\mathbb{E}[Y^3] = \mu_J^3 + 3\mu_J\sigma_J^2$, $\mathbb{E}[Y^4] = \mu_J^4 + 6\mu_J^2\sigma_J^2 + 3\sigma_J^4$ (moments of a Normal variable)
3. Substitute into the table below; skewness $c_3/c_2^{3/2} = -2.07$, excess kurtosis $c_4/c_2^2 = 17.6$

	Diffusion	Jumps	Total
$c_2 (\times 10^{-4})$	$\sigma^2 \Delta = 0.893$	$\lambda \Delta \mathbb{E}[Y^2] = 0.496$	1.389 (35.7% jumps)
$c_3 (\times 10^{-6})$	0	-3.393	-3.393
$c_4 (\times 10^{-7})$	0	3.399	3.399

- **Interpretation:** five modest jumps a year, one day in 50, make daily returns strongly non-Normal; the diffusion adds variance but no higher cumulants

A7 [Solved]: Solution (2/2) — Information on λ 

1. $\ell(\lambda) = N_T \ln \lambda - \lambda T + \text{const}$, $\ell'' = -N_T/\lambda^2$, so $I(\lambda) = \mathbb{E}[N_T]/\lambda^2 = T/\lambda$ and $\text{s.e.}(\hat{\lambda}) = \sqrt{\lambda/T}$
 2. Scenario 1, $\lambda = 5$: s.e. $\sqrt{5/36.7} = 0.37$, a relative s.e. of 7.4% (red curve)
 3. Scenario 2, the lecture fit $\hat{\lambda} = 101.9$: observed jumps would give s.e. 1.67, but the MLE s.e. is 10.5, 6.3 times larger
- **Interpretation:** the jumps are latent: the Poisson mixture must separate them from diffusive moves, which destroys most of the information about λ ; the left panel shows how little of the density the jumps occupy

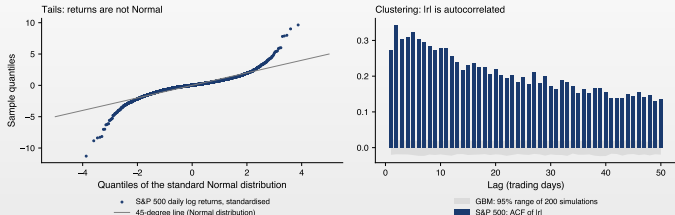
B2 [Solved]: GBM on the S&P 500 — Task

- **Question:** which GBM parameter do 36 years of daily data pin down, and which GBM assumption fails?
- **Data:** S&P 500 closes, GSPC.INDX (field `close`), 2 January 1990 – 18 September 2026, weekdays, stale quotes removed; $n = 9\,245$ daily log returns, $T = 36.7$ years, $\Delta t = 1/252$
- **Tasks**
 1. Compute $\hat{\sigma}^2 = \frac{1}{n\Delta t} \sum (r_t - \bar{r})^2$, $\hat{m} = \bar{r}/\Delta t$, $\hat{\mu} = \hat{m} + \hat{\sigma}^2/2$, with $\text{s.e.}(\hat{\sigma}) = \hat{\sigma}/\sqrt{2n}$, $\text{s.e.}(\hat{m}) = \hat{\sigma}/\sqrt{T}$ and a 95% CI for m .
 2. Test Normality with Jarque–Bera and independence with Ljung–Box on $|r_t|$ (20 lags), both at 5%, and plot the QQ plot and the ACF of $|r_t|$.
 3. Compute the variance ratio of non-overlapping 20-day sums (blocks from the first return; the incomplete last block dropped) to 20 times the daily variance.
 4. Recompute the s.e. without GBM: $\hat{\sigma}\sqrt{(k+2)/(4n)}$ for independent returns with excess kurtosis k , and a moving-block bootstrap (blocks of 63 days, 999 samples) [Künsch \(1989\)](#).
- **Report:** estimates with GBM and bootstrap s.e. (percentage points), test statistics with critical values and decisions, two charts
- **Interpretation:** which parameter is known well and which badly? Can $\hat{\sigma}$ still be used for option pricing?
- Notebook: section B2

B2 [Solved]: Solution (1/3) — Estimates Step by Step

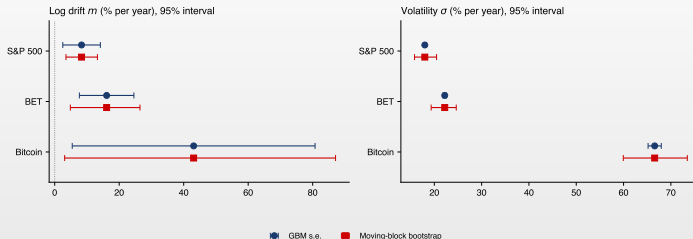
1. $n = 9\,245$, $T = 36.7$: $\hat{\sigma} = 18.0\%$, s.e. $= \hat{\sigma}/\sqrt{2n} = 0.13$ p.p.; $\hat{m} = 8.3\%$, s.e. $= \hat{\sigma}/\sqrt{T} = 3.0$ p.p.
2. $\hat{\mu} = \hat{m} + \hat{\sigma}^2/2 = 10.0\%$; GBM 95% CI for m : $[2.5\%, 14.2\%]$; ± 1 point would need $(1.96 \hat{\sigma}/0.01)^2 = 1\,243$ years
3. Without GBM, s.e. of $\hat{\sigma}$: 0.34 p.p. with kurtosis $k = 10.9$, 1.15 p.p. by block bootstrap, 8.7 times the GBM value; m : block s.e. 2.5 p.p., CI $[3.5\%, 13.3\%]$ (chart on slide 3/3)

B2 [Solved]: Solution (2/3) — Diagnostics



1. QQ plot: tails far beyond the 45-degree line; Jarque–Bera 45 805 against the 5% critical value 5.99: Normality rejected
 2. ACF of $|r_t|$: 0.27 at lag 1, far outside the GBM range; Ljung–Box 13 381 against $\chi^2_{20,0.95} = 31.4$: independence rejected
 3. Variance ratio 0.75, below 1: a descriptive sign of mild mean reversion in 20-day sums; no uncertainty attached, so no test
- **Interpretation:** both stylised facts of Chapter 1 reject GBM; $\hat{\sigma}$ is an average of a time-varying volatility, a benchmark for pricing, not a calibration

B2 [Solved]: Solution (3/3) — GBM versus Block-Bootstrap Uncertainty

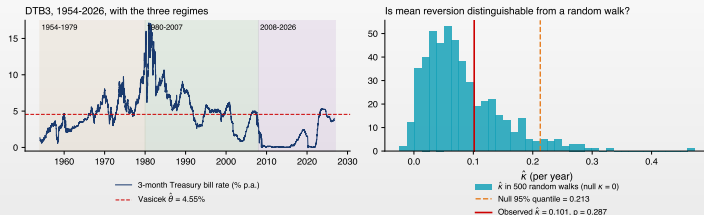


- Blue: GBM intervals; red: moving-block bootstrap; S&P 500 (B2), BET and Bitcoin (B3), each on its own sample and calendar
- Volatility: the bootstrap interval is several times wider than the GBM one, yet still narrow relative to $\hat{\sigma}$; drift: both intervals are wide, and Bitcoin's barely excludes zero
- Interpretation:** σ is known well even without GBM, the drift badly under any method: a long-span problem

B4 [Solved]: Vasicek and the Bias of $\hat{\kappa}$ — Task

- **Question:** does the 3-month Treasury bill rate really revert to a mean, once the bias of $\hat{\kappa}$ is accounted for?
- **Data:** FRED series DTB3 (3-month US Treasury bill rate), daily, 4 January 1954 – 18 September 2026, divided by 100 (decimals); $n = 18\,170$ observations, $T = 72.1$ years, $\Delta t = 1/252$
- **Tasks**
 1. Regress x_{t+1} on x_t by OLS (the exact AR(1) of A9), recover κ , θ , σ with delta-method s.e., a 95% CI for κ and the half-life [Vasicek \(1977\)](#).
 2. Bias experiment: simulate 500 exact OU paths of the same length, started from the stationary law, with the estimates as the truth, and re-estimate κ on each.
 3. Compute the bias (mean estimate minus truth), compare it with $4/T$ [Tang & Chen \(2009\)](#), and give the bias-corrected κ with an approximate interval.
 4. Null experiment: simulate 500 random walks ($\kappa = 0$, no drift) of the same length from x_0 , with the estimated residual s.d.; p = share of null $\hat{\kappa}$ at or above the observed one, with the $+1$ rule.
- **Report:** estimates with s.e., the two simulated distributions and the corrected values
- **Interpretation:** is mean reversion significant before and after the correction? What does the bias do to long-maturity rate risk?
- Notebook: section B4

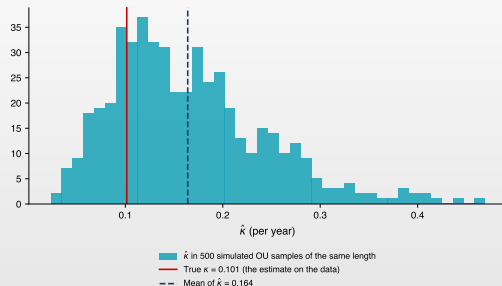
B4 [Solved]: Solution (1/2) — Estimates and the Random-Walk Null



1. $\hat{b} = 0.99960$, so $\hat{\kappa} = -\ln \hat{b} / \Delta t = 0.101$ (s.e. 0.051), Wald 95% CI [0.001, 0.202]; $\hat{\theta} = 4.55\%$ (s.e. 1.55), $\hat{\sigma} = 1.33\%$, half-life 6.8 years
 2. Null (right panel): under a random walk $\hat{\kappa}$ averages 0.082, not 0, and its 95% quantile is 0.213; the observed 0.101 has $p = 0.287$: a random walk is not rejected
- **Interpretation:** the Wald CI barely excludes zero because it ignores the skewed small-sample law of $\hat{\kappa}$; the three regimes (left) also give very different local estimates

B4 [Solved]: Solution (2/2) — Bias of $\hat{\kappa}$ and Correction

1. 500 exact OU samples of 72.1 years with $\kappa = 0.101$: mean $\hat{\kappa} = 0.164$ (s.d. 0.078), bias 0.063, close to $4/T = 0.055$; 78% of the estimates exceed the truth
 2. Corrected $\kappa = 0.101 - 0.063 = 0.039$, approximate interval $0.039 \pm 1.96 \cdot 0.078 = [-0.114, 0.192]$: it contains zero, so no finite half-life can be reported with confidence
- **Interpretation:** the data are compatible with slow reversion but cannot exclude a random walk; with $\kappa^Q = \kappa$, the upward bias makes Vasicek understate long-maturity rate risk



C2 [Proposed]: Critique an AI Answer — Task

- **Question:** a student asked an assistant: “S&P 500 as a GBM with $\mu = 8\%$, $\sigma = 20\%$: what is $P(S_1 < S_0)$? Check by simulation.”

□ **The answer (numbered lines)**

1. $dS = \mu S dt + \sigma S dW$
2. Chain rule: $d \ln S = dS/S = \mu dt + \sigma dW$, so $\ln(S_T/S_0) \sim N(\mu T, \sigma^2 T)$
3. $P(S_T < S_0) = \Phi(-\mu\sqrt{T}/\sigma) = \Phi(-0.40) = 0.345$
4. $\mathbb{E}[S_T] = S_0 e^{\mu T}$: on average 8.3% higher
5. “The simulation gives 0.345 and confirms the formula”

```
1 import numpy as np
2 mu, sigma, T, n, N = 0.08, 0.20, 1.0, 252, 20_000
3 dt = T / n
4 rng = np.random.default_rng(1)
5 Z = rng.standard_normal((N, n))
6 lnS = np.cumsum((mu - 0.5*sigma**2)*dt
7               + sigma*dt*Z, axis=1)
8 print((lnS[:, -1] < 0).mean())
```

□ **Tasks**

1. The text and the code contain three errors; find them by line number.
2. For each, give a check that detects it: a formula, a test or a run of the code.
3. Give the corrected formula, code line and number, with a Monte Carlo s.e.

- **Report:** a table: error, line, check, correction; log it in AI_ERRORS.md. Model: A4, B2

B3 [Proposed]: GBM on BET and Bitcoin — Task

- **Question:** which market is further from GBM, and is Bitcoin's drift significantly positive?
- **Data:** BET (BET, close, 2000–2026, weekdays, $\Delta t = 1/252$, $n = 6\,670$); Bitcoin (BTC-USD.CC, close, 18 September 2014 – 18 September 2026, 7 days a week, $\Delta t = 1/365$, $n = 4\,384$); different sample periods
- **Tasks**
 1. For each market, compute $\hat{\sigma}$ and $\hat{m}$ with GBM and block-bootstrap 95% CIs (blocks of 63 observations) and the years needed for ± 1 point on m .
 2. Compute Jarque–Bera, Ljung–Box on $|r_t|$ (20 lags, 5%) and the 20-observation variance ratio, and draw the QQ and ACF charts.
 3. State that 20 and 63 observations are 4 weeks and a quarter for BET but under 3 weeks and 2 months for Bitcoin.
- **Report:** one table (rows: $\hat{\sigma}$, $\hat{m}$ with both CIs, years, kurtosis, JB, Ljung–Box, lag-1 ACF, variance ratio; columns: BET, Bitcoin) and the charts
- **Interpretation:** what can a variance ratio above 1 for BET suggest, and what else would you need to call it thin trading?
- **Model:** B2; **Notebook:** section B3

Submission Checklist and Project Menu

Submit (team repository)

- A9, A4, A7: the answers with every step
- B2, B4, B3 (and B1, B6 after the lecture): one table and one chart each, every estimate with its s.e. or CI, one interpretation sentence
- C2: the error table and the corrected code with its printed output
- Code that reruns from the data files; declare AI use in `AI_USE.md` and the errors caught in `AI_ERRORS.md`

Before you submit, check

- the $-\sigma^2/2$ term (price drift versus log drift)
- units: annual parameters with the right Δt (Bitcoin: 1/365)
- no χ^2 critical values for a parameter on the boundary
- $\hat{\kappa}$ from a short span is biased upwards
- Project menu: C1 (which model fits BET and Bitcoin) or C3 (rough volatility of SPY), in the extension packs

After the Lecture: Two Solved Exercises

- B1 and B6 use tools that [Chapter 11](#) introduces after this seminar; work through them once you have seen the lecture
- B1: the Euler–Maruyama and Milstein discretisations of an SDE (stochastic differential equation) and their strong and weak orders of convergence [Higham \(2001\)](#)
 - ▶ Euler–Maruyama: $X_{t+\Delta} = X_t + a(X_t)\Delta + b(X_t)\Delta W$; Milstein adds $\frac{1}{2}bb'(X_t)[(\Delta W)^2 - \Delta]$, the Itô correction of the primer
- B6: the Merton likelihood as a Poisson mixture of Normal densities, fitted by maximum likelihood, and a parametric-bootstrap likelihood-ratio test against GBM [Merton \(1976\)](#)
 - ▶ Why a bootstrap: under GBM $\lambda = 0$ lies on the boundary and μ_J, σ_J are not identified, so the χ^2 limit does not apply
- Both are [Solved]: reproduce the tables and charts from the notebook sections B1 and B6

B1 [Solved]: Orders of Convergence — Task

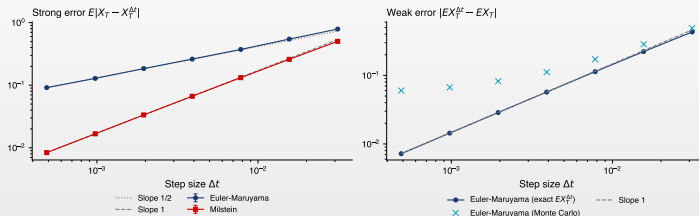
- **Question:** how fast do the Euler–Maruyama and Milstein schemes converge, and what limits a Monte Carlo check?
- **Data:** $dX = 2X dt + X dW$, $X_0 = 1$, $T = 1$, exact solution $X_T = \exp(1.5 + W_T)$ [Higham \(2001\)](#); 20,000 paths on a fine grid of 2^{11} steps; a coarse increment is the sum of $2^{11} \Delta t$ fine increments, so every scheme and the exact value use the same path
- **Tasks**
 1. For $\Delta t = 2^{-11}, \dots, 2^{-5}$, store the path-by-step array of absolute errors $|X_T - X_T^{\Delta t}|$ for both schemes and average over paths.
 2. Regress $\ln(\text{error})$ on $\ln \Delta t$ (the global slope); get its s.e. and 95% CI by resampling whole paths, all step sizes jointly (1,000 samples).
 3. Compute the weak error of Euler–Maruyama exactly, from $\mathbb{E}[X_T^{\Delta t}] = (1 + 2\Delta t)^{1/\Delta t}$, and by the Monte Carlo mean; estimate both slopes.
- **Report:** a table of the four slopes with bootstrap s.e. and CI, and the log-log chart
- **Interpretation:** do the slopes agree with the orders 1/2 and 1? Why does the Monte Carlo weak error give a wrong slope?
- Notebook: section B1

B1 [Solved]: Solution (1/2) — Slopes Step by Step

	Global slope	s.e.	95% CI
Strong, Euler–Maruyama	0.518	0.003	[0.512, 0.525]
Strong, Milstein	0.986	0.001	[0.984, 0.987]
Weak, exact mean	0.985	–	deterministic
Weak, Monte Carlo	0.511	0.269	[0.247, 1.289]

1. Global strong slopes are close to $1/2$ and 1; their CIs measure Monte Carlo noise only and can exclude $1/2$ and 1, because coarse steps are pre-asymptotic
2. The local slopes between the two finest steps, 0.492 (Euler) and 0.998 (Milstein), are the ones that approach the limiting orders
3. Weak, exact mean: a deterministic slope close to 1 (local 0.998), with no sampling interval

B1 [Solved]: Solution (2/2) — The Chart and the Monte Carlo Floor

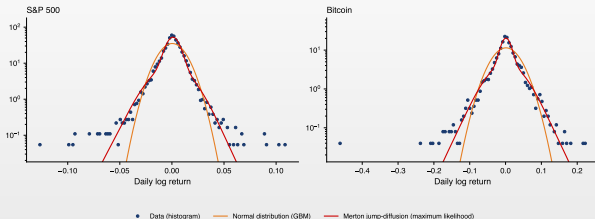


1. Left: strong errors with reference slopes $1/2$ and 1 ; halving Δt halves the Milstein error but cuts Euler only by $1 - 2^{-1/2} = 29\%$
 2. Right: the Monte Carlo s.e. of the mean, $e^2 \sqrt{e - 1} / \sqrt{20\,000} = 0.068$, exceeds the exact weak error 0.0072 at 2^{-11} , so the crosses flatten
 3. For this direct-mean estimator, resolving the error at 95% needs $(1.96 \cdot 9.69 / 0.0072)^2 \approx 6\,937\,381$ paths
- **Interpretation:** for expected values the Monte Carlo sample size, not the time step, is the binding constraint

B6 [Solved]: Merton and a Bootstrap Likelihood-Ratio Test — Task

- ▣ **Question:** do daily S&P 500 returns reject GBM in favour of the Merton jump-diffusion?
- ▣ **Data:** S&P 500 closes, GSPC.INDX (field `close`), 2 January 1990 – 18 September 2026, weekdays, stale quotes removed; $n = 9\,245$ daily log returns, $T = 36.7$ years, $\Delta t = 1/252$
- ▣ **Tasks**
 1. Fit $(m, \sigma, \lambda, \mu_J, \sigma_J)$ by maximising $\sum_t \log \sum_{k=0}^{12} p_k \varphi(r_t; m\Delta t + k\mu_J, \sigma^2\Delta t + k\sigma_J^2)$, $p_k =$ Poisson weights (log-sum-exp), on $\sigma \geq 5\%$, from six starting points, with Hessian s.e. [Merton \(1976\)](#)
 2. Compute the likelihood-ratio statistic $LR = 2(\ell_{\text{Merton}} - \ell_{\text{GBM}})$ on this restricted space.
 3. Simulate 200 GBM samples of the same length from the fitted GBM, refit both models with the same restriction and starts, and compute the 95% critical value and $p = (1 + \#\{LR^* \geq LR\})/201$.
- ▣ **Report:** all five parameters with s.e., LR, the bootstrap and χ_3^2 critical values, p
- ▣ **Interpretation:** why is χ_3^2 not valid here, and does rejecting GBM prove that prices jump?
- ▣ Notebook: section B6

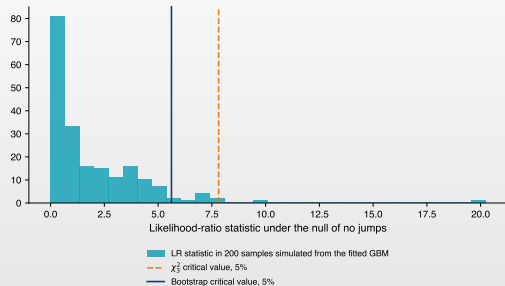
B6 [Solved]: Solution (1/2) — Fitted Parameters and Density



1. $\hat{m} = 24.8\%$ (2.6), $\hat{\sigma} = 9.44\%$ (0.32), $\hat{\lambda} = 101.9$ (10.5) per year, $\hat{\mu}_J = -0.162\%$ (0.034), $\hat{\sigma}_J = 1.44\%$ (0.06); s.e. in brackets
 2. Implied skewness -0.31 (data -0.36), excess kurtosis 3.71 (data 10.9): the fitted density (chart, left) follows the centre, but two thirds of the kurtosis is missed
- **Interpretation:** about 100 small “jumps” a year stand in for changing volatility, not for rare crashes

B6 [Solved]: Solution (2/2) — Bootstrap Null Distribution and Decision

1. $LR = 2401$ ($\sigma \geq 5\%$ not binding); bootstrap 95% critical value 5.62 against $\chi^2_{3,0.95} = 7.81$
 2. 0 of the 200 bootstrap values exceed the LR, so $\hat{p} = (1 + 0)/201 = 0.0050$, the smallest value this design can give (95% Monte Carlo upper bound 0.015): GBM is rejected at 5%
 3. Under H_0 , $\lambda = 0$ is on the boundary and μ_J, σ_J are unidentified: Wilks' theorem fails, hence the bootstrap
- **Interpretation:** the test rejects i.i.d. Normal returns; it does not identify discontinuous paths, since stochastic volatility would also reject GBM (lecture, mini-case)



Extension Packs: What Each One Needs

Exercise	Question	Type	Model
A1	GBM under P and Q : which probability forecasts, which prices?	Solved	–
A2	Siegel's paradox: can both sides expect to gain?	Proposed	A1
A3	the Vasicek bond and yield curve by Feynman–Kac	Solved	–
A5	CIR: moments, transition law, Feller condition	Solved	–
A6	what does the VIX identify in Heston?	Proposed	A5
A8	rare, large jumps: can crash risk be learnt?	Proposed	A7
B4 Extended	exact CIR, Euler and CKLS on DTB3	Solved	–
B6 Extended	is the Merton likelihood bounded?	Solved	–
B5, B7	OU for the VIX; jumps in Bitcoin	Proposed	B4; B6
B8, B8 Extended	Heston from the VIX; the affine map and realised variance	Proposed	B4, B6; A6
B9	nonparametric diffusion of the T-bill rate	Proposed	B4 Extended
C1, C3	project seeds: models for BET and Bitcoin; rough and persistent SPY volatility	Proposed	B2, B6, B8; B1, B4



A1 [Solved]: GBM under P and Q (Girsanov) — Task

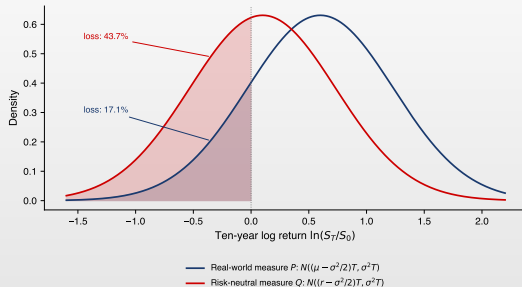
- ▣ **Question:** which probability of a ten-year loss is a forecast, and which one is a pricing weight?
- ▣ **Inputs:** under the real-world measure P , $dS = \mu S dt + \sigma S dW$ with $\mu = 8\%$, $\sigma = 20\%$; risk-free rate $r = 3\%$; $T = 10$ years; Q is the measure under which $e^{-rt}S_t$ is a martingale; Φ : standard Normal distribution function; no data
- ▣ **Tasks**
 1. Find the market price of risk θ that removes the drift of $e^{-rt}S_t$ and write the density $Z_T = dQ/dP$ [Girsanov \(1960\)](#).
 2. Using $\mathbb{E}[e^{aW_T}] = e^{a^2T/2}$, show that $\mathbb{E}^P[Z_T S_T] = S_0 e^{rT}$; state whether σ changes under Q .
 3. Compute $P(S_{10} < S_0)$ and $Q(S_{10} < S_0)$ separately, from the lognormal law under each measure.
- ▣ **Report:** θ , Z_T , $\mathbb{E}^Q[S_{10}]/S_0$ and the two probabilities
- ▣ Notebook: section A1

A1 [Solved]: Solution (1/2) — Girsanov Step by Step

1. Product rule: $d(e^{-rt}S_t) = e^{-rt}S_t[(\mu - r)dt + \sigma dW_t]$; with $dW = d\tilde{W} - \theta dt$ the drift is $\mu - r - \sigma\theta = 0$ for $\theta = (0.08 - 0.03)/0.20 = 0.25$
2. Girsanov with constant θ (Novikov holds): $Z_T = \exp(-\theta W_T - \frac{1}{2}\theta^2 T)$
3. $\mathbb{E}^P[Z_T S_T] = S_0 e^{(\mu - \sigma^2/2 - \theta^2/2)T} \mathbb{E}[e^{(\sigma - \theta)W_T}] = S_0 e^{(\mu - \sigma^2/2 - \theta^2/2 + (\sigma - \theta)^2/2)T} = S_0 e^{(\mu - \sigma\theta)T} = S_0 e^{rT}$, so $\mathbb{E}^Q[S_{10}]/S_0 = e^{0.3} = 1.350$
4. Under Q , $dS = rS dt + \sigma S d\tilde{W}$: σ is unchanged, since $[\ln S]_T = \sigma^2 T$ is a property of the path

A1 [Solved]: Solution (2/2) — Two Loss Probabilities

1. Log drifts $m_P = \mu - \sigma^2/2 = 6\%$ and $m_Q = r - \sigma^2/2 = 1\%$; $\ln(S_{10}/S_0) \sim N(10m, 0.4)$ under each measure
 2. $P(S_{10} < S_0) = \Phi(-0.06\sqrt{10}/0.20) = \Phi(-0.949) = 17.1\%$; $Q(S_{10} < S_0) = \Phi(-0.01\sqrt{10}/0.20) = \Phi(-0.158) = 43.7\%$ (shaded areas)
- **Interpretation:** the P -probability forecasts a loss; the Q -probability is a pricing weight that embeds the risk premium and so overweights bad outcomes; the two densities differ only by their drift



 MFM_ch11_seminar

A2 [Proposed]: Siegel's Paradox and the Numéraire — Task

- ▣ **Question:** if the EUR/RON rate (lei per euro) follows GBM, is the drift of RON/EUR (euros per lei) just the opposite?
- ▣ **Inputs:** X_t = lei per euro, GBM with $\mu = 2\%$, $\sigma = 4.4\%$ under P ; $Y_t = 1/X_t$; for step 4: constant rates r_{RON} , r_{EUR} (symbolic), money-market accounts $B_t^{RON} = e^{r_{RON}t}$ in lei and $B_t^{EUR} = e^{r_{EUR}t}$ in euro, worth $X_t B_t^{EUR}$ in lei; given numbers, no data
- ▣ **Tasks**
 1. Apply Itô's lemma to $f(x) = 1/x$ and write dY_t .
 2. Compute the percentage drift of Y to four decimals and compare it with $-\mu$; then compare the log drifts of X and Y .
 3. Show that $F = X_0 e^{(r_{RON} - r_{EUR})T} = \mathbb{E}^{Q^{RON}}[X_T]$, using that $X_t B_t^{EUR} / B_t^{RON}$ is a Q^{RON} -martingale.
 4. Optional: with the numéraire density $dQ^{EUR}/dQ^{RON} = X_T B_T^{EUR} / (X_0 B_T^{RON})$, show that $\mathbb{E}^{Q^{EUR}}[1/X_T] = 1/F$.
- ▣ **Report:** the drifts and the two expectations
- ▣ **Interpretation:** why is there no arbitrage when both a RON and a euro investor seem to expect a gain?
- ▣ **Model:** A1; Notebook: section A2

A3 [Solved]: The Vasicek Bond by Feynman–Kac — Task

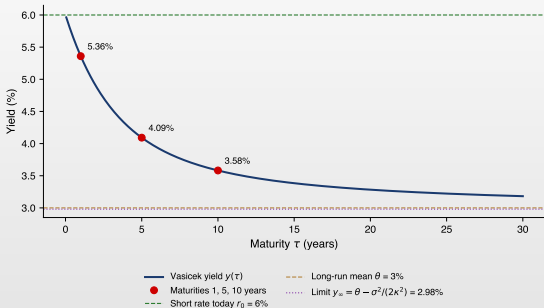
- ▣ **Question:** what yield curve does a Vasicek short rate imply today?
- ▣ **Inputs:** under Q , $dr = \kappa(\theta - r)dt + \sigma d\tilde{W}$, $\kappa = 0.5$, $\theta = 3\%$, $\sigma = 1\%$ (risk premium already in θ), $r_0 = 6\%$ [Vasicek \(1977\)](#); a zero-coupon bond pays 1 at T ; $p(t, r; T)$ is its price, $\tau = T - t$, and a prime denotes $d/d\tau$
- ▣ **Tasks**
 1. Write the Feynman–Kac PDE and terminal condition for $p(t, r; T) = \mathbb{E}^Q[e^{-\int_t^T r_s ds} \mid r_t = r]$ [Kac \(1949\)](#).
 2. Insert $p = \exp(A(\tau) - B(\tau)r)$, collect terms in r and solve for B and $A = \int_0^\tau [-\kappa\theta B(s) + \sigma^2 B(s)^2/2] ds$.
 3. Compute $y(\tau) = (Br_0 - A)/\tau$ for $\tau = 1, 5, 10$, the limit y_∞ , and plot the curve.
- ▣ **Report:** $B(5)$, $A(5)$, three yields, y_∞ and the chart
- ▣ **Interpretation:** why does the curve slope down, and why is y_∞ below θ ?
- ▣ **Notebook:** section A3

A3 [Solved]: Solution (1/2) — From the PDE to Two Ordinary Differential Equations

1. Feynman–Kac: $p_t + \kappa(\theta - r)p_r + \frac{1}{2}\sigma^2 p_{rr} - rp = 0$, $p(T, r; T) = 1$
2. With $p = e^{A - Br}$ and $\partial_t = -d/d\tau$: $p_t = (-A' + B'r)p$, $p_r = -Bp$, $p_{rr} = B^2p$; dividing by p :
 $-A' + B'r - \kappa(\theta - r)B + \frac{1}{2}\sigma^2 B^2 - r = 0$ for every r
3. Terms in r : $B' = 1 - \kappa B$, $B(0) = 0$, so $B(\tau) = (1 - e^{-\kappa\tau})/\kappa$
4. Constants: $A' = -\kappa\theta B + \frac{1}{2}\sigma^2 B^2$; integrating with $\int_0^\tau B = (\tau - B)/\kappa$ and $\int_0^\tau B^2 = (\tau - B)/\kappa^2 - B^2/(2\kappa)$:
$$A(\tau) = \left(\theta - \frac{\sigma^2}{2\kappa^2}\right)(B - \tau) - \frac{\sigma^2 B^2}{4\kappa}$$

A3 [Solved]: Solution (2/2) — Numbers and the Yield Curve

1. $B(5) = (1 - e^{-2 \cdot 5})/0.5 = 1.836$;
 $A(5) = (0.03 - 0.0002)(1.836 - 5) - 0.0001 \cdot 1.836^2/2 = -0.0945$
 2. $y(1) = 5.36\%$, $y(5) = 4.09\%$, $y(10) = 3.58\%$;
 $y_\infty = \theta - \sigma^2/(2\kappa^2) = 2.98\%$
- **Interpretation:** $r_0 = 6\%$ is above θ , so expected rates fall and the curve slopes down; $y_\infty < \theta$ by convexity (Jensen): rate uncertainty raises bond prices and lowers long yields



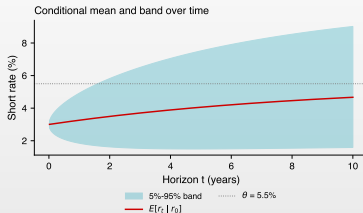
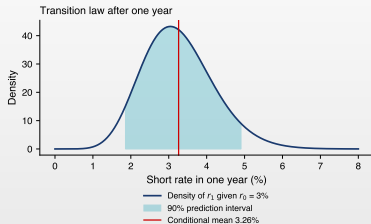
A5 [Solved]: CIR: Moments, Transition Law and the Feller Condition — Task

- **Question:** where can a CIR short rate be in one year, and can it reach zero?
- **Inputs:** $dr = \kappa(\theta - r)dt + \sigma\sqrt{r}dW$, $\kappa = 0.11$, $\theta = 5.5\%$, $\sigma = 0.056$ (close to the exact CIR fit of B4 Extended) [Cox, Ingersoll & Ross \(1985\)](#); $r_0 = 3\%$, $t = 1$; transition $2c r_t \mid r_0 \sim \chi^2_\nu(\delta)$ with $c = 2\kappa/(\sigma^2(1 - e^{-\kappa t}))$, $\nu = 4\kappa\theta/\sigma^2$, noncentrality $\delta = 2c r_0 e^{-\kappa t}$
- **Tasks**
 1. Apply Itô's lemma to $e^{\kappa t} r_t$ and derive $\mathbb{E}[r_t \mid r_0]$.
 2. Apply Itô's lemma to r_t^2 , solve the ordinary differential equation for $m_2(t) = \mathbb{E}[r_t^2]$ with an integrating factor and derive $\text{Var}(r_t \mid r_0)$.
 3. Compute c , ν , δ and the 5% and 95% quantiles of r_1 (`scipy.stats.ncx2.ppf`, then divide by $2c$).
 4. Show that $2\kappa\theta \geq \sigma^2 \Leftrightarrow \nu \geq 2$ and evaluate the Feller ratio.
- **Report:** mean, s.d., c , ν , δ , the 90% conditional prediction interval (not a parameter CI), the Feller ratio
- **Interpretation:** why is the interval asymmetric around the mean?
- Notebook: section A5

A5 [Solved]: Solution (1/2) — Conditional Moments

1. $d(e^{\kappa t} r) = \kappa \theta e^{\kappa t} dt + \sigma e^{\kappa t} \sqrt{r} dW$; the stochastic integral has mean zero ($\mathbb{E}[r_s] < \infty$), so $\mathbb{E}[r_t] = \theta + (r_0 - \theta)e^{-\kappa t}$
2. $d(r^2) = 2r dr + \sigma^2 r dt$, so $m'_2 = (2\kappa\theta + \sigma^2)\mathbb{E}[r_t] - 2\kappa m_2$; multiply by $e^{2\kappa t}$ and integrate:
 $m_2(t) = r_0^2 e^{-2\kappa t} + (2\kappa\theta + \sigma^2) \int_0^t e^{-2\kappa(t-s)} \mathbb{E}[r_s] ds$
3. Subtracting $\mathbb{E}[r_t]^2$: $\text{Var} = r_0 \frac{\sigma^2}{\kappa} (e^{-\kappa t} - e^{-2\kappa t}) + \frac{\theta \sigma^2}{2\kappa} (1 - e^{-\kappa t})^2$
4. $\mathbb{E}[r_1] = 0.055 + (0.03 - 0.055)e^{-0.11} = 3.26\%$; s.d. 0.94%

A5 [Solved]: Solution (2/2) — Transition Law and Feller



1. $c = 673$, $\nu = 7.72$, $\delta = 36.2$ (not the jump intensity); 90% prediction interval [1.85%, 4.92%]
2. $\nu = 2 \cdot (2\kappa\theta/\sigma^2)$, so $\nu \geq 2 \Leftrightarrow 2\kappa\theta \geq \sigma^2$; Feller ratio 3.86: zero is never reached

□ **Interpretation:** volatility $\sigma\sqrt{r}$ grows with the level, so the law is skewed to the right and the band widens upwards (right panel); negative rates are impossible, unlike Vasicek

A6 [Proposed]: The VIX under Heston — Task

- ▣ **Question:** which Heston parameters does a regression on the squared VIX recover, and under which measure?
- ▣ **Inputs:** instantaneous variance v_t with $dv = \kappa^Q(\theta^Q - v)dt + \xi_v\sqrt{v}d\tilde{W}^{(2)}$ under Q and κ^P, θ^P under P (same ξ_v); $y_t = (\text{VIX}_t/100)^2 = \frac{1}{\tau}\mathbb{E}^Q[\int_t^{t+\tau} v_s ds \mid \mathcal{F}_t]$, $\tau = 30/365$; $\kappa^Q = 5$, $\theta^Q = 0.044$; a CIR fit to y gives $\hat{\xi}_y = 0.56$
- ▣ **Tasks**
 1. Using A5 under Q , integrate $\mathbb{E}^Q[v_s \mid v_t]$ over $[t, t + \tau]$ and show $y_t = a + b v_t$; compute b and a .
 2. From $dy = b dv$, write the diffusion of y and the ratio ξ_y/ξ_v as a function of v ; evaluate the local correction $\hat{\xi}_y/b$ at $v = \theta^Q$.
 3. Derive the P -dynamics of y and state which of κ^P, θ^P, ρ a historical regression on y recovers, and what ρ additionally needs.
- ▣ **Report:** a, b , the local correction and the chart of ξ_y/ξ_v
- ▣ **Interpretation:** is 0.56 too high or too low for ξ_v , and what is needed to estimate κ^Q ?
- ▣ **Model:** A5; **Notebook:** section A6

A8 [Proposed]: Rare, Large Jumps — Task

- ▣ **Question:** can the intensity of rare crashes be estimated from one index history?
- ▣ **Inputs:** the A7 model with $\sigma = 15\%$ but $\lambda = 0.5$ per year, $\mu_J = -10\%$, $\sigma_J = 5\%$ (A7: $\lambda = 5$, $\mu_J = -3\%$, $\sigma_J = 4\%$); $\Delta = 1/252$; $T = 36.7$ years; lecture MLE for the S&P 500: $\hat{\lambda} = 101.9$
- ▣ **Tasks**
 1. Fill the A7 table (diffusion and jump parts of c_2 , c_3 , c_4) for the new parameters and compute skewness and excess kurtosis.
 2. Compute the probability of no jump in 1 and in 10 years.
 3. With every crash observed, compute the s.e. of $\hat{\lambda}$ over 36.7 years and the years needed for a 20% relative s.e.
- ▣ **Report:** the moment table, the two probabilities, the s.e. and the years
- ▣ **Interpretation:** what does the lecture MLE capture instead, and is the problem only precision or also model uncertainty?
- ▣ **Model:** A7; **Notebook:** section A8

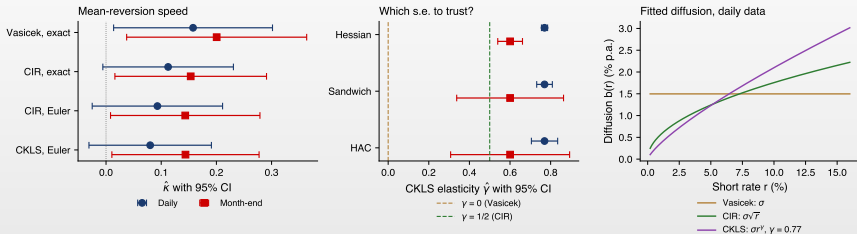
B4 Extended [Solved]: Exact CIR, Euler and CKLS — Task

- ▣ **Question:** which volatility specification do short-rate data prefer, and which test can be trusted?
- ▣ **Inputs:** DTB3, 4 January 1954 – 31 December 2007 (after 2008: 13 exact zeros, 8 negative quotes, outside the CIR and CKLS domains), daily ($n = 13\,487$, $\Delta = 1/252$) and month-end ($n = 647$, $\Delta = 1/12$); $\kappa, \theta, \sigma > 0$
- ▣ **Tasks**
 1. Maximise the exact CIR log-likelihood $\sum_t [\ln(2c) + \ln f_{\chi^2_{\nu}(\delta_t)}(2c r_{t+\Delta})]$ (the $2c$ is the Jacobian, A5) and the Euler Gaussian pseudo-likelihood; compare $\hat{\kappa}$, $\hat{\theta}$ with Hessian s.e.
 2. Fit CKLS $dr = \kappa(\theta - r)dt + \sigma r^\gamma dW$ by the Euler quasi-likelihood; s.e. of $\hat{\gamma}$ from H^{-1} , the sandwich $H^{-1}\Omega H^{-1}$ with Ω the score outer product, and HAC Ω (Bartlett, $\lfloor 4(n/100)^{2/9} \rfloor$ lags) [Chan, Karolyi, Longstaff & Sanders \(1992\)](#); [Newey & West \(1987\)](#).
 3. Test $\gamma = 0$ and $\gamma = 1/2$ by the LR of restricted against unrestricted Euler objectives and by HAC Wald tests, at 5% (3.84), at both frequencies.
- ▣ **Report:** a table of $\hat{\kappa}$ (s.e.), $\hat{\theta}$; $\hat{\gamma}$ with three s.e.; the test statistics; the chart
- ▣ **Interpretation:** why do the LR and the Hessian s.e. over-reject?
- ▣ Notebook: section B4 Extended

B4 Extended [Solved]: Solution (1/2) — Estimates

Model	Daily		Month-end	
	$\hat{\kappa}$ (s.e.)	$\hat{\theta}$ (%)	$\hat{\kappa}$ (s.e.)	$\hat{\theta}$ (%)
Vasicek, exact	0.158 (0.073)	5.39	0.201 (0.083)	5.36
CIR, exact	0.113 (0.060)	5.48	0.153 (0.070)	5.43
CIR, Euler	0.093 (0.060)	5.55	0.144 (0.069)	5.45
CKLS, Euler	0.080 (0.057)	5.84	0.144 (0.068)	5.46

1. Exact and Euler CIR differ by less than one s.e. of $\hat{\kappa}$ at both frequencies: discretisation error is small next to long-span uncertainty
2. All $\hat{\kappa}$ lie between 0.08 and 0.20, with s.e. 0.06–0.08: no likelihood pins down the speed (left panel of the chart, next slide)

B4 Extended [Solved]: Solution (2/2) — Tests on γ 

1. $\hat{\gamma} = 0.77$ daily (s.e.: Hessian 0.008, sandwich 0.020, HAC 0.033, 11 lags); 0.60 monthly (0.031, 0.134, 0.149, 6 lags)
2. $\gamma = 0$: LR 7926.6 and 285.6, HAC Wald 545.1 and 16.1: Vasicek rejected at both frequencies
3. $\gamma = 1/2$: daily LR 1139.34 and HAC Wald 67.01 reject (22.13 with 55 lags); monthly LR 10.07 rejects, HAC Wald 0.45 does not

□ **Interpretation:** volatility rises with the level; omitted clustering makes the scores heteroskedastic and autocorrelated, so trust the HAC Wald; Chan, Karolyi, Longstaff & Sanders (1992) found $\hat{\gamma} \approx 1.5$ on 1964–1989

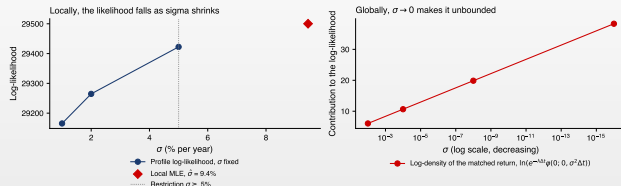
B5 [Proposed]: Mean Reversion of the VIX — Task

- ▣ **Question:** how fast do volatility spikes fade, and is one OU factor enough?
- ▣ **Data:** VIX.INDX, close, 1990–2026, weekdays; $x_t = \ln VIX_t$, $\Delta t = 1/252$, $n = 9\,283$ transitions
- ▣ **Tasks**
 1. Fit the OU process by the exact AR(1) (B4, step 1) and give $\hat{\kappa}$, its s.e. and the stationary median level $e^{\hat{\theta}}$.
 2. Convert $\hat{\kappa}$ into a half-life in trading days, $252 \ln 2 / \hat{\kappa}$, with the interval $[252 \ln 2 / \kappa_{\text{upper}}, 252 \ln 2 / \kappa_{\text{lower}}]$ (both ends positive).
 3. Estimate the bias with 300 simulated OU paths, and compare the sample ACF with $\hat{b}^k$ up to lag 120 on a chart.
- ▣ **Report:** $\hat{\kappa}$, $e^{\hat{\theta}}$, half-life with interval, bias, the ACF chart
- ▣ **Interpretation:** at which horizons does one OU factor fail, and what extension would you try?
- ▣ **Model:** B4; **Notebook:** section B5

B6 Extended [Solved]: Is the Merton Likelihood Bounded? — Task

- ▣ **Question:** what does the reported Merton MLE mean if the likelihood has no maximum?
- ▣ **Inputs:** the S&P 500 returns and the B6 fit ($\hat{\sigma} = 9.4\%$); the fitting helper of B6 with σ fixed [Honoré \(1998\)](#)
- ▣ **Tasks**
 1. Compute the profile log-likelihood with σ fixed at 5%, 2% and 1%, refitting the other four parameters.
 2. Set $m\Delta t$ equal to one observed return and let $\sigma \rightarrow 0$: compute that return's term $\ln(e^{-\lambda\Delta t}\varphi(0; 0, \sigma^2\Delta t))$ at $\sigma = 10^{-2}, 10^{-4}, 10^{-8}, 10^{-16}$.
- ▣ **Report:** the profile table and the singular sequence, with the chart
- ▣ **Interpretation:** is the likelihood bounded, and why can numerical profiles alone not settle it?
- ▣ **Notebook:** section B6 Extended

B6 Extended [Solved]: Solution — Local Profile and Global Singularity



σ (fixed)	log-likelihood	$\hat{\lambda}$ per year	$\hat{\sigma}_J$ (%)
9.4% (MLE)	29 500	101.9	1.44
5%	29 422	302	0.93
2%	29 265	555	0.70
1%	29 165	710	0.62

1. Locally the profile falls as σ shrinks while $\hat{\lambda}$ grows (left panel); the MLE satisfies $\sigma \geq 5\%$
 2. Globally the matched return's term is 6.0, 10.7, 19.9, 38.3 at $\sigma = 10^{-2}, 10^{-4}, 10^{-8}, 10^{-16}$, while the other returns keep finite densities from the jump components: $\sup = +\infty$ (right panel)
- **Interpretation:** the argument, not the numbers, proves unboundedness; the MLE is an interior local maximum, so the LR test of B6 is defined on $\sigma \geq 5\%$ for data and bootstrap alike

B7 [Proposed]: Jumps in Bitcoin — Task

- ▣ **Question:** how often does Bitcoin jump, by a fitted mixture and by a detection test?
- ▣ **Data:** BTC-USD.CC, close, 18 September 2014 – 18 September 2026, 7 days a week, $\Delta t = 1/365$; Lee–Mykland:
 $L_t = r_t / \hat{\sigma}_t$, $\hat{\sigma}_t^2 = \frac{1}{K-2} \sum_{j=t-K+2}^{t-1} |r_j| |r_{j-1}|$, $K = 16$; statistic $(|L_t| - C_n)/S_n$ against $-\ln(-\ln(0.99))$ (lecture formulas for C_n, S_n) [Lee & Mykland \(2008\)](#)
- ▣ **Tasks**
 1. Fit Merton as in B6 and report all five parameters with s.e., the implied excess kurtosis and the jump share of variance.
 2. Apply the test: the first $K - 1$ days have no local scale and are not tested, $n =$ number of tested days; count the flagged days per year and their mean absolute size, and plot returns and statistic.
- ▣ **Report:** the Merton table and the Lee–Mykland counts side by side, with the chart
- ▣ **Interpretation:** why do $\hat{\lambda}$ and the count differ by two orders of magnitude? Why is neither the risk-neutral intensity that prices a one-week option?
- ▣ **Model:** B6; **Notebook:** section B7

B8 [Proposed]: Heston Parameters from the VIX — Task

- ▣ **Question:** what do S&P 500 and VIX data say about κ , θ , ξ and ρ , and are they stable?
- ▣ **Data:** S&P 500 (as in B2) and VIX.INDX joined on common days, 1990–2026, then returns; $y_t = (\text{VIX}_t/100)^2$ in place of v_t , $\Delta t = 1/252$ [Heston \(1993\)](#)
- ▣ **Tasks**
 1. Regress $(y_{t+\Delta} - y_t)/\sqrt{y_t}$ on $\Delta t/\sqrt{y_t}$ and $-\Delta t\sqrt{y_t}$ without an intercept: $\kappa = \beta_2$, $\theta = \beta_1/\beta_2$, $\xi = s_e/\sqrt{\Delta t}$; give OLS and HAC s.e. and a 95% CI for κ .
 2. Estimate ρ as the correlation of $r_{t+1}/\sqrt{y_t}$ with the regression residual over the same interval, with a Fisher-z CI (assumes i.i.d. bivariate Normal shocks), and the Feller ratio $2\kappa\theta/\xi^2$.
 3. Repeat on 1990–2007 and 2008–2026.
- ▣ **Report:** a table with the full sample and the two halves
- ▣ **Interpretation:** why is $\sqrt{\theta}$ from the VIX above the realised volatility?
- ▣ **Model:** B4, B6; Notebook: section B8

B8 Extended [Proposed]: The Affine Map and Realised Variance — Task

- ▣ **Question:** does ρ depend on which variance proxy we use?
- ▣ **Inputs:** as in B8, plus the SPY 5-minute realised variance (RV) of [Chapter 9](#) (total, overnight included, in squared percent): annualise by $\text{RV} \times 10^{-4} / \Delta t$; join S&P 500, VIX and RV prices on common days first (October 2020 – September 2026), then returns
- ▣ **Tasks**
 1. With $\kappa^Q = 5$ and A6, compute b and the local correction $\hat{\xi}/b$, and state which estimates need no correction.
 2. On the common days, estimate ρ twice (VIX and RV in place of v_t), with the same return timing and Fisher-z 95% CIs; draw both scatter plots.
- ▣ **Report:** the corrected ξ , the two correlations with CIs, the chart
- ▣ **Interpretation:** why is ρ from RV much closer to zero, and which one would you use in a Heston model?
- ▣ **Model:** A6, B8; **Notebook:** section B8 Extended

B9 [Proposed]: Nonparametric Diffusion of the T-Bill Rate — Task

- ▣ **Question:** which parametric diffusion, if any, matches the data at every rate level?
- ▣ **Data:** DTB3 daily, 1954–2007, $\Delta = 1/252$; the fits of B4 Extended [Bandi & Phillips \(2003\)](#); grid of 60 levels between the 2% and 98% quantiles; Gaussian kernel, $h = 1.06 s_r n^{-1/5}$; weights $w_t(x) = K((r_t - x)/h) / \sum_s K((r_s - x)/h)$
- ▣ **Tasks**
 1. Compute $\hat{a}(x) = \sum_t w_t \Delta r_t / \Delta$ and $\hat{b}^2(x) = \sum_t w_t (\Delta r_t)^2 / \Delta$ with s.e. $\sqrt{\sum_t w_t^2 (y_t - \hat{m})^2}$; bands for b are the square roots of the b^2 bands, truncated at zero.
 2. Overlay the Vasicek, CIR and CKLS diffusions and compute the share of levels inside the pointwise band.
 3. Repeat with $h/2$ and $2h$; report the log-slope of $\hat{b}(r)$ and the share of levels where the drift band contains zero.
- ▣ **Report:** the chart and a table by bandwidth
- ▣ **Interpretation:** can the drift be distinguished from zero? What do the bands ignore?
- ▣ **Model:** B4 Extended; Notebook: section B9

C1 [Proposed]: Which Continuous-Time Model Fits BET and Bitcoin? — Task

- **Question:** which stylised facts does each fitted model reproduce in markets without an implied-volatility index?
- **Data:** BET and Bitcoin daily log returns (Data Used); variance proxy $RV_t = \frac{1}{21\Delta t} \sum_{j=0}^{20} r_{t-j}^2$, an overlapping window: for i.i.d. returns its lag- k autocorrelation is $(21 - k)/21$, so persistence is partly built in (control: simulated ACF 0.952 at lag 1)
- **Tasks**
 1. Start with one market: fit GBM (B2), Merton (B6) and Heston by the B8 regression with RV_t in place of v_t .
 2. Simulate 100 samples of the data length per model (Heston: Euler with full truncation, daily step, θ set to the realised variance, $v_0 = \theta$, 500 burn-in steps, fixed seed) and compute excess kurtosis, the Hill statistic $\left[\frac{1}{k} \sum_{i \leq k} \ln(L_{(i)}/L_{(k+1)}) \right]^{-1}$ of the losses $L = -r$ ($k = 5\%$ of returns), skewness and $\sum_{k=1}^{20}$ ACF of $|r_t|$.
 3. Report $q =$ share of simulated statistics at or below the data value (e.g. $q = 0.97$: data above 97 of 100 samples), a descriptive check since parameters are not refitted.
- **Report:** medians with 90% ranges, the q table, the charts; optional: Bates (Heston plus jumps) as a separate advanced project
- **Interpretation:** which features does no model reproduce? Is Bitcoin a distinct class [Pele et al. \(2023\)](#)?
- Model: B2, B6, B8; Notebook: section C1

C3 [Proposed]: Is SPY Volatility Rough and Persistent? Stage 1 — Task

- **Question:** is daily SPY log-volatility rough ($\alpha < 0$) and consistent with long memory ($\beta < 1$)?
- **Data:** SPY 5-minute bars, complete 09:30–16:00 New York sessions (78 bars); first return of a session $\log(\text{close/open})$ of its first bar, then close to close, no overnight return; lecture case study [Bennedsen, Lunde & Pakkanen \(2022\)](#)
- **Tasks**
 1. Compute daily bipower variation $BV_t = \frac{\pi}{2} \frac{M}{M-1} \sum_{j=2}^M |r_j| |r_{j-1}|$, $M = 78$ [Barndorff-Nielsen & Shephard \(2004\)](#), and $x_t = \log \hat{\sigma}_t = \frac{1}{2} \log BV_t$; plot the series.
 2. Estimate α by OLS of $\log \hat{\gamma}_2(k)$ on $\log k$, $k = 1, \dots, 6$, $\hat{\alpha} = (\text{slope} - 1)/2$, and by NLLS $\hat{\gamma}_2(k) = b_0 + b_1 k^{2\alpha+1}$ averaged over $m = 10, \dots, 20$ lags.
 3. Fit the Cauchy ACF $\rho(h) = (1 + h^{2\alpha+1})^{-\beta/(2\alpha+1)}$ and an exponential (OU) ACF to the first $\lceil n^{1/3} \rceil$ lags; plot both against the sample ACF to lag 100.
- **Report:** $\hat{\alpha}_{\text{OLS}}$, $\hat{\alpha}_{\text{NLLS}}^*$, $\hat{\beta}$, the OU half-life, three charts; reference values in the lecture
- **Interpretation:** is “consistent with long memory” the same as “has long memory”? How robust are your estimates to the lags used?
- **Model:** B1, B4; Notebook: section C3

C3 [Proposed]: Stages 2 and 3 (Advanced, Optional)

- ▣ **Stage 2, intraday:** $\Delta = 30$ and 65 minutes ($M = 6$ and 13 returns per bin), $\hat{\sigma}_t^2 = BV_t/\Delta$ divided by a seasonal factor
 - ▶ Seasonal factor: flexible Fourier form for each bin's volatility [Andersen & Bollerslev \(1997\)](#), scaled so that its square averages 1 over the session, estimated from past data only; plot the intraday profile before and after
 - ▶ Lags h and λ are in steps of Δ : compare with Table 3, Panel A of the paper only in the same unit; bars instead of every trade mean no pre-averaging and more noise
- ▣ **Stage 3, forecasting:** at $\Delta = 65$ minutes, forecast the variance summed over $h = 1, 5, 20$ steps with log-HAR (heterogeneous autoregressive model for log volatility) [Corsi \(2009\)](#), RFSV [Gatheral, Jaisson & Rosenbaum \(2018\)](#) and the Cauchy model, rolling window of 200 observations, only data up to each origin
 - ▶ Losses MSE and QLIKE, 90% MCS [Hansen, Lunde & Nason \(2011\)](#); plot realised against forecast variance and cumulative QLIKE differences; the MCS does not rank the models it contains
- ▣ Assess robustness and reproducibility, not whether the paper's ranges are matched

References (1/3)

- ▣ Andersen, T.G., Bollerslev, T. (1997). Intraday periodicity and volatility persistence in financial markets. *Journal of Empirical Finance*, 4(2–3), 115–158.
- ▣ Bandi, F.M., Phillips, P.C.B. (2003). Fully nonparametric estimation of scalar diffusion models. *Econometrica*, 71(1), 241–283.
- ▣ Barndorff-Nielsen, O.E., Shephard, N. (2004). Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1), 1–37.
- ▣ Bennedsen, M., Lunde, A., Pakkanen, M.S. (2022). Decoupling the short- and long-term behavior of stochastic volatility. *Journal of Financial Econometrics*, 20(5), 961–1006.
- ▣ Bollerslev, T., Zhou, H. (2002). Estimating stochastic volatility diffusion using conditional moments of integrated volatility. *Journal of Econometrics*, 109(1), 33–65.
- ▣ Borak, S., Härdle, W.K., López-Cabrera, B. (2013). *Statistics of Financial Markets: Exercises and Solutions*, 2nd ed. Springer.
- ▣ Chan, K.C., Karolyi, G.A., Longstaff, F.A., Sanders, A.B. (1992). An empirical comparison of alternative models of the short-term interest rate. *Journal of Finance*, 47(3), 1209–1227.
- ▣ Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- ▣ Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- ▣ Cox, J.C., Ingersoll, J.E., Ross, S.A. (1985). A theory of the term structure of interest rates. *Econometrica*, 53(2), 385–407.

References (2/3)

- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Gatheral, J., Jaisson, T., Rosenbaum, M. (2018). Volatility is rough. *Quantitative Finance*, 18(6), 933–949.
- ▣ Girsanov, I.V. (1960). On transforming a certain class of stochastic processes by absolutely continuous substitution of measures. *Theory of Probability and Its Applications*, 5(3), 285–301.
- ▣ Hansen, P.R., Lunde, A., Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.
- ▣ Heston, S.L. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies*, 6(2), 327–343.
- ▣ Higham, D.J. (2001). An algorithmic introduction to numerical simulation of stochastic differential equations. *SIAM Review*, 43(3), 525–546.
- ▣ Honoré, P. (1998). Pitfalls in estimating jump-diffusion models. Working paper, SSRN.
- ▣ Itô, K. (1944). Stochastic integral. *Proceedings of the Imperial Academy*, 20(8), 519–524.
- ▣ Kac, M. (1949). On distributions of certain Wiener functionals. *Transactions of the American Mathematical Society*, 65(1), 1–13.

References (3/3)

- ▣ Künsch, H.R. (1989). The jackknife and the bootstrap for general stationary observations. *Annals of Statistics*, 17(3), 1217–1241.
- ▣ Lee, S.S., Mykland, P.A. (2008). Jumps in financial markets: a new nonparametric test and jump dynamics. *Review of Financial Studies*, 21(6), 2535–2563.
- ▣ Merton, R.C. (1976). Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3(1–2), 125–144.
- ▣ Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *European Journal of Finance*, 29(10), 1064–1105.
- ▣ Samuelson, P.A. (1965). Rational theory of warrant pricing. *Industrial Management Review*, 6(2), 13–39; reprinted in *Henry P. McKean Jr. Selecta* (2015), Birkhäuser.
- ▣ Tang, C.Y., Chen, S.X. (2009). Parameter estimation and bias correction for diffusion processes. *Journal of Econometrics*, 149(1), 65–81.
- ▣ Uhlenbeck, G.E., Ornstein, L.S. (1930). On the theory of the Brownian motion. *Physical Review*, 36(5), 823–841.
- ▣ Vasicek, O. (1977). An equilibrium characterization of the term structure. *Journal of Financial Economics*, 5(2), 177–188.